

Project Ideas

6.S953: Topics in Error-Correcting Codes for Computer Science · Fall 2026

Projects are completed in groups of **two or three**. You can choose either a paper reading project or a research project. Each topic below introduces a problem and the key papers for a reading project. Where included, a **Research remark** suggests directions to explore. You may also propose a topic outside this list!

Format and timeline reminders — The course project is worth 50% of the course grade.

Due date	Deliverable	Course grade
October 9 at 6:00 PM	Topic and group selection	5%
October 16 at 6:00 PM	One-page project proposal and plan	10%
November 13 at 6:00 PM	One-page progress report	10%
December 7 and 9	Project presentations	15%
December 11	Eight- to ten-page project report or paper	10%

Questions and project discussions: Contact either instructor with questions about project topics, scope, or requirements.

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Module 1. Foundations

1. Improvements to MRRW bounds: OpenAI and Alrabiah–Guruswami

How large can a binary code of relative distance δ be? Lecture 4 discusses the first McEliece–Rodemich–Rumsey–Welch (MRRW) bound $R \leq H_2(1/2 - \sqrt{\delta(1-\delta)})$, where H_2 is binary entropy [Goy26]. The MRRW bounds, proved in 1977, remained the best general asymptotic upper bounds for binary codes for nearly fifty years.

Chapter 2 of OpenAI breaks this barrier by representing codewords as subspaces whose overlaps depend on Hamming distance, then using linear algebra to bound the number of codewords [Ope26]. Concurrently, Alrabiah–Guruswami’s method based on classical–quantum channels strictly improves both MRRW bounds for every $0 < \delta < 1/2$ [AG26]. OpenAI and Alrabiah–Guruswami provide comparable improvements through different proof techniques.

2. List decoding alphabet size bounds

In class, we showed that completely random codes of rate $1 - H_q(p) - \varepsilon$ are $(p, O(1/\varepsilon))$ list-decodable. Can random linear codes match this? That question remains open, with progress in both the small- and large-alphabet regimes. Kumar–Mon prove list size $O(qH_q(p)/\varepsilon)$ for random linear codes, for all p, q, ε , and connect this to hashing and Furstenberg sets [KM26]. Goyal–Guruswami achieve $O(1/\varepsilon)$ lists near the Singleton radius over alphabets of size $\exp(\widetilde{O}(1/\varepsilon))$ [GG26]; further work sharpens asymptotic list sizes for fixed alphabets [YZ26, Sil26].

Research remark. Improve the alphabet-size bounds for codes meeting the ε -relaxed generalized Singleton bound. For small alphabets, improve the list sizes achieved, for example by saving the $O(q)$ additional factor from [KM26].

3. LCC and LDC lower bounds via Kikuchi matrices and rainbow cycles

Locally correctable codes (LCCs) and locally decodable codes (LDCs) allow recovery of any codeword bit or message bit, respectively, with high probability by querying only constantly many locations in a corrupted codeword. How small can the block length of a good LCC or LDC be? Recent work has given new lower bounds.

Kikuchi-matrix methods give near-cubic LDC and exponential linear-LCC lower bounds for three queries; rainbow-cycle estimates yield near-tight bounds for binary linear 3-query LCCs. Choose one of the two techniques and read the corresponding papers from [AGKM23, KM23, ABS⁺23, AG24].

Module 2. Interactive coding theory

4. Explicit tree-code constructions

A tree code encodes a message as it arrives: each output symbol depends only on the message prefix seen so far, and two messages have well-separated encodings on the suffix after they first diverge. This protects online communication, but explicit constructions with a small alphabet remain difficult. Cohen–Haeupler–Schulman give a polylogarithmic alphabet, Ben Yaacov–Cohen–Yankovitz improve it to sublogarithmic, and Yankovitz’s recent note gives a related construction using elementary binary shifts. Read [CHS18, BYCY21, Yan25].

5. Error correction with a constant number of feedback rounds

How much noiseless feedback is needed to beat the binary unique-decoding limit of $1/4$? Gupta–Guruswami–Zhang achieve $1/3 - \varepsilon$ bit-flip resilience using only $O_\varepsilon(1)$ feedback rounds and $O_\varepsilon(\log k)$ feedback bits for a k -bit message, replacing feedback after every transmitted bit. A matching $\Omega(\log k)$ feedback lower bound holds beyond $1/4$ resilience. Read [GGZ22].

Research remark. Improve the rate and computational efficiency while retaining $1/3 - \varepsilon$ resilience and $O_\varepsilon(1)$ feedback rounds. Determine the tradeoff between rate, feedback, and the gap ε to optimal resilience.

6. Deterministic binary interactive coding at optimal resilience

Interactive coding simulates a two-party protocol despite adversarial corruption in both directions. Gupta–Zhang give a randomized simulation tolerating $1/6 - \varepsilon$ bit flips with constant rate and efficient computation, using layered codes; $1/6$ is the optimal threshold in the fixed-length, fixed-speaking-order binary model. Read [GZ22a].

Research remark. Can we remove the randomness? Aim for a deterministic two-party interactive coding scheme achieving $1/6 - \varepsilon$ error resilience.

7. Interactive error-correcting codes: resilience and communication

In an interactive error-correcting code (iECC), only Alice has a k -bit message; Bob can reply, and corruptions in both directions count against the total communication. For erasures, known constructions tolerate $3/5 - \varepsilon$ with $O_\varepsilon(k^2)$ communication [GKZ21] or $6/11 - \varepsilon$ with $O_\varepsilon(k)$ [GZ22b], versus the upper bound $2/3$ [GKZ21]. For bit flips, constant-rate codes achieve $1/4 + 10^{-5}$ [EKSZ22], while the upper bound is $13/47$ [GZ23].

Research remark. Tighten the gaps between achievable resilience and impossibility bounds for erasures or bit flips. A concrete efficiency target is $3/5 - \varepsilon$ erasure resilience with linear communication; a separate target is to raise the achievable resilience toward $2/3$ for erasures or $13/47$ for bit flips.

Module 3. Code constructions

8. c^3 locally testable codes

Locally testable codes (LTCs) admit a tester that queries only a few coordinates to distinguish codewords from words far from the code. The c^3 challenge is to get constant rate, constant relative distance, and constant query complexity, all with constant soundness. Two breakthroughs achieve all of this: Dinur–Evra–Livne–Lubotzky–Mozes construct such codes using left-right Cayley complexes and high-dimensional expanders; independently, Panteleev–Kalachev provide an alternate construction using lifted products. Read [DEL⁺21, DEL⁺22, PK22].

9. Constructing ε -balanced codes via expander walks

We want explicit binary linear codes whose nonzero codewords have relative weight in $[(1 - \varepsilon)/2, (1 + \varepsilon)/2]$. The Gilbert–Varshamov (GV) random-coding benchmark is rate $\Omega(\varepsilon^2)$; see Lecture 4 [Goy26]. Ta-Shma’s breakthrough work used expander walks to achieve a construction of rate $\varepsilon^{2+o(1)}$, and recently Hsieh–Mohanty–Zhang provided an arguably simpler construction also achieving rate $\varepsilon^{2+o(1)}$ [TS17, HMZ26].

10. Explicit constructions of subspace designs

How can we explicitly construct the subspace designs from Module 4 over small alphabets? Algebraic constructions and the Alon–Edmonds–Luby (AEL) distance-amplification method achieve strong subspace-design guarantees but have large alphabets [GK13, GGH26]. Function-field constructions offer another approach [GXY17, GRSZ26].

Research remark. Seek an explicit construction beyond folded Reed–Solomon (FRS) codes, univariate multiplicity codes, and the AEL approach, for example through folded algebraic-geometric codes or function fields. Another target is a smaller alphabet in the AEL construction. Preserve the strong subspace-design property needed for the intended applications.

Module 4. Subspace designs and local coordinate-wise linear properties

11. Low-randomness constructions and subspace designs

Random folded linear codes are subspace designs with small alphabet, but sampling them can use many random bits. Putterman–Pyne showed that methods based on expander walks give capacity-achieving list-decodable codes with less randomness, while Goyal–Guruswami–Sun–Wootters showed that random low-density parity-check (LDPC) codes also have similar properties. These suggest routes to pseudorandom subspace-design constructions. Read [PP23, GGSW26].

Research remark. Construct pseudorandom subspace-design codes with alphabet size $2^{O(1/\varepsilon^2)}$ and profile $\tau(1/\varepsilon) \leq R + \varepsilon$ using $o(n)$ random bits. More broadly, develop pseudorandom constructions as an intermediate step toward explicit ones.

12. Line decoding and proximity gaps of subspace-design codes

Line decoding generalizes list decoding from a single received word to an affine line $u_0 + \alpha u_1$: it studies the structure of codewords close to points on that line. Proximity gaps turn sufficiently many nearby points on a line into a guarantee that the whole line is close to the code, a property used in proof systems. Goyal–Guruswami establish strong guarantees for subspace-design codes, and Goyal–Guruswami–Sun–Wootters obtain matching guarantees for random code ensembles. Read [GG25a, GGSW26].

Research remark. What is the minimum a for which a random linear code of rate R over a sufficiently large alphabet is $(1 - R - \varepsilon, a, 3)$ line-decodable? That is, how many codewords within relative distance $1 - R - \varepsilon$ of a line $u_0 + \alpha u_1$ force three of them to be collinear? The known sufficient bound is $O(1/\varepsilon^2)$ and the necessary bound is $\Omega(1/\varepsilon)$; the conjectured threshold is $\Theta(1/\varepsilon)$.

Alternatively, find applications of line decoding and proximity gaps outside proof systems and cryptography.

13. List recovery, structure, and Brascamp–Lieb inequalities

A code C is (δ, ℓ, L) list-recoverable if for all sets $S_1, \dots, S_n \subseteq \Sigma$ with $|S_i| = \ell$, we have that $|\{c \in C \mid \Delta(c, S_1 \times S_2 \times \dots \times S_n) \leq \delta n\}| \leq L$. We know that at $\delta = 1 - R - \varepsilon$, any linear code has list size $L \geq \Omega(\ell^{R/\varepsilon+1})$, and our best known combinatorial upper bound, $(\ell/(R + \varepsilon))^{R/\varepsilon+1}$, is close to this lower bound but not tight; it follows from discrete Brascamp–Lieb inequalities [BCDZ25a].

Research remark. Sharpen the characterization of list-recovery outputs and obtain tight list-size bounds. Start with alternate Brascamp–Lieb formulations, or investigate the conjecture in the final section of Goyal–Guruswami’s structure paper [GG25b] using techniques based on Brascamp–Lieb inequalities.

14. Optimality of subspace designs based on folded Reed–Solomon codes

How strong can subspace designs be? Section 6 of Brakensiek–Chen–Dhar–Zhang proves an optimality bound over algebraically closed fields, matching subspace designs based on folded Reed–Solomon (FRS) codes, but over finite fields, this bound is not tight. Read [BCDZ25b].

Research remark. Extend the optimality bound beyond algebraically closed fields to show that FRS-based subspace designs are essentially optimal over finite fields. Getting an $o_q(1)$ error term, tending to zero with field size q , would already be exciting progress.

15. Local properties of random LDPC codes: field-independent sparsity

Low-density parity-check (LDPC) codes are linear codes defined by sparse parity checks. Work of Mosheiff–Resch–Ron–Zewi–Silas–Wootters and Goyal–Guruswami–Sun–Wootters shows that local coordinate-wise linear (LCL) properties of random linear codes also hold for random LDPC codes with constant sparsity. The required parity-check sparsity currently depends on the field size q . Read [MRRZ⁺19, GGSW26].

Research remark. Remove the dependence on the field size q from the sparsity required for random LDPC codes to inherit LCL properties. Replace the condition $s > s_0(\eta, r, R, q)$ by $s > s_0(\eta, r, R)$, uniformly over q , where η is the threshold gap, r the locality, and R the rate. The starting point is to identify and eliminate the q -dependent losses in the proof of Mosheiff–Resch–Ron–Zewi–Silas–Wootters.

Module 5. Algorithms

16. List decoding Reed–Solomon codes over prime fields

We want every Reed–Solomon codeword within relative distance $1 - R - \varepsilon$ of a received word. Going beyond the Johnson radius $1 - \sqrt{R}$ is difficult even combinatorially: Guruswami–Rudra and Ben-Sasson–Kopparty–Radhakrishnan exhibit superpolynomial lists in extension-field, low-rate regimes [GR06, BSKR10]. Those constructions exploit extension-field structure and do not apply to prime fields.

For fixed R, ε and a sufficiently large prime field of size $q = \Theta_{R,\varepsilon}(n)$, Brakensiek–Chen–Putterman–Zhang–Zheng give deterministic polynomial-time decoding for every set of evaluation points, with list size $n^{O_{R,\varepsilon}(1)}$ [BCP⁺26]. Thus even arbitrary evaluation sets admit polynomial-size lists near capacity in this regime! The exponent depends on both R and ε .

Research remark. There are two natural directions to explore:

1. Improve the combinatorial list-size upper or lower bounds for prime-field Reed–Solomon codes: is dependence on n inherent, or can the list size depend only on R, ε ? The generalized Singleton bound forces $\Omega_R(1/\varepsilon)$ [ST20].
2. Speed up the decoder using fast interpolation and root-finding ideas from [Ale05, GHKS24, GHKS25, Kop15, CHK26, KTS22]. Discuss with Rohan for more specific suggestions here.

17. List decoding AEL-based subspace-design codes

Alon–Edmonds–Luby (AEL) distance amplification encodes each symbol of an outer code with an inner code, then redistributes and groups the resulting symbols along an expander graph to increase relative distance. Srivastava–Tulsiani give near-linear-time decoding of these codes via graph regularity lemmas [ST25].

Research remark. Develop a Guruswami–Wang-style [GW12] algorithm for an appropriate AEL construction that finds, in near-linear time, a low-dimensional space containing every nearby codeword. Aim for better parameter dependence than explicitly recovering the codewords in the Srivastava–Tulsiani algorithm; stronger assumptions such as the subspace-design property may be useful.

18. Decoding dual-BCH codes

Dual-BCH codes encode a polynomial g over $\mathbb{F}_q$, where $q = 2^m$, by the binary trace values $(\text{Tr}(g(\alpha)))_{\alpha \in \mathbb{F}_q}$. Write $d = \deg g$ and restrict g to odd-degree monomials. Earlier decoders correcting a constant fraction of errors ran in $d^{O(d)}$ poly(log q) time, giving polynomial time in q only for $d = O(\log q / \log \log q)$. For larger d , the standard Reed–Muller approach corrected only $O(q/d)$ errors.

Kopparty gives a poly(q)-time decoder correcting fewer than $(1/8 - \varepsilon)q$ errors for $d \leq \varepsilon\sqrt{q}$, where $0 < \varepsilon < 1/8$ [Kop26]. This takes constant-fraction error correction from polylogarithmic dimension all the way to dimension on the order of $\sqrt{q} \log q$ for fixed ε . The algorithm combines Stepanov’s approach to Weil bounds with Berlekamp–Welch interpolation using pseudopolynomials.

Research remark. Improve the decoding radius toward $1/4 - \varepsilon$, the relative unique-decoding radius guaranteed by the Weil bound. Identify the losses in the interpolation argument that prevent reaching this threshold.

Module 6. Applications

19. The PCP theorem without composition

The probabilistically checkable proof (PCP) theorem is one of the great achievements of theoretical computer science: every problem in NP has polynomial-length proofs that a randomized verifier can check by reading only constantly many bits. A central technique in proving this theorem is proof composition, which is analogous to code concatenation. Is proof composition necessary?

A recent result of Amireddy–Behera–Srinivasan–Sudan–Willumsgaard proves the PCP theorem without any proof composition [ABS⁺26]!

20. Polynomial-factor hardness of CVP

Given a binary linear code and a received word, the nearest-codeword problem asks for a codeword minimizing Hamming distance. The closest vector problem (CVP) is the analogous problem for lattices, using Euclidean distance. How well can we approximate these minimum distances in polynomial time?

The previous best NP-hardness factor for Euclidean CVP was the almost-polynomial $n^{c/\log \log n}$ of Dinur–Kindler–Raz–Safra (DKRS), for an absolute constant $c > 0$ [DKRS03]. Its exponent tends to zero; obtaining hardness for a fixed positive power of n remained open.

Chapter 7 of the OpenAI report proves NP-hardness of approximation within factors $n^{1/200}$ for binary nearest codeword and $n^{1/400}$ for Euclidean CVP, where n is the block length or lattice dimension, respectively [Ope26]. This replaces the vanishing exponent in the CVP hardness factor with a fixed positive constant. The proof gives a direct deterministic reduction from 3SAT using Reed–Solomon encodings and polynomial reconstruction.

21. Cryptographic pseudorandom error-correcting codes

Can an error-correcting code look random? Pseudorandom codes combine error correction with codewords that are computationally indistinguishable from random strings. The initial constructions enable robust watermarking and steganography; later work strengthens robustness and security and develops new constructions from permuted puzzles. Read [CG24, GG24, AAC⁺24, CGG⁺25].

Research remark. Find new applications of cryptographic pseudorandom codes outside proof systems and cryptography. Identify a problem where simultaneous error correction and computational indistinguishability give a useful algorithmic or complexity-theoretic consequence, and determine the security and robustness guarantees it requires.

References

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